# Emily Steiner

PhD Candidate, Stanford University | 3D/4D Computer Vision

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## Research Interests

I study how machines can learn from temporal patterns in complex, long-term observations to understand changing objects and environments. My research spans 3D/4D scene understanding, grounded spatiotemporal representations, and vision-language models, with applications in embodied AI, robotics, and construction.

## Education

**Stanford University** Sept 2023–2028 (expected)
*MS–PhD Candidate, Electrical Engineering* | Advisor: Prof. Iro Armeni, Gradient Spaces Lab.

**University of Waterloo** Sept 2018– June 2023
*BASc, Honours Mechatronics Engineering, Computing Option, Co-op*
With Distinction; Dean's Honour List. Cumulative average: 95.6%.

## Publications

**ReScene4D: Temporally Consistent Semantic Instance Segmentation of Evolving Indoor 3D Scenes.**
Emily Steiner, Jianhao Zheng, Henry Howard-Jenkins, Chris Xie, Iro Armeni.
**CVPR 2026.** Paper | Project | Code

## Relevant Experience

**Stanford University, Gradient Spaces Lab** Apr 2024–present
*Graduate Research Assistant | Advisor: Prof. Iro Armeni*

- Developed **ReScene4D**, a query-based transformer whose joint spatiotemporal feature encoding and query decoding improve temporal identity consistency over independent 3D segmentation with ad hoc matching.
- Introduced t-mAP to measure segmentation and temporal identity consistency; showed that sharing features across revisits also improves single-scan segmentation.

**Waymo** Jun–Sep 2026
*Technical Intern | Mentor: Chris Ying*

- Fine-tuned the Gemma4 vision-language model with LoRA in JAX for fine-grained video action recognition, achieving +15 pp mAcc over zero-shot; evaluated temporal context for long-tail scenarios.
- Designed a labelling policy and compared hierarchical with single-label classification; the hierarchical task improved prediction while eliciting structured reasoning without chain-of-thought supervision.
- Built an end-to-end video annotation pipeline integrating multi-target tracking, temporal label propagation, and dataset curation for data-efficient training.

**Stanford University, Computational Imaging Lab** Jan–Apr 2024
*PhD Rotation Student | Advisor: Prof. Gordon Wetzstein*

- Built an audio simulator comparing head- and gaze-directed beamforming for hearing aids.

## Selected Projects

**Crowdsourcing 4D Data from Evolving Living Spaces** Winter 2026

- Built a 3D/4D data pipeline using Swift/ARKit LiDAR scans, CLIP/Concerto validation, and Django/AWS.

**Probing the Object Awareness of 3D LLMs** Fall 2024

- Identified density-induced hallucination and object detection bottlenecks through 3D LLM perturbation tests.

**Open Scene Understanding for Construction Analysis** Spring 2024

- Lifted GroundingDINO, SAM, and CLIP semantics into 3D meshes for zero-shot construction monitoring.

## Honours, Awards & Grants

**TomKat Graduate Fellowship for Translational Research** 2025–2027
**CIFE Seed Grant** | Research staff; PI: Prof. Iro Armeni 2025–2026
Queryable spatiotemporal 4D representations for construction progress monitoring.
**Ansys Design Analysis Competition, 1st place, Waterloo** 2023
Drag Reduction System Automation capstone with Williams Racing Formula 1.
**Co-op Student of the Year, Honourable Mention, Waterloo** 2022
**First in Class Engineering Scholarship, Waterloo** 2022, 2021, 2019

## Teaching & Academic Service

**3DV 2027** | Student Organizer Current
**Stanford CEE 147C/247C** | Teaching Assistant 2026–present

- Teaching assistant for Computer Vision for the Built Environment; leading the Winter 2027 course redesign.

**Defining What Construction AI Needs to See** | Lead Workshop Organizer Sept 2026

- Led 70+ industry participants in evaluating 3D job-site scans and defining scene-understanding priorities.

**Computer Vision for the Built World** | Workshop Organizer CVPR 2026
**Nothing Stands Still** | Challenge Organizer ICRA 2025; CVPR 2026
**Stanford Women in Electrical Engineering** 2024–2026

- Faculty Liaison (2024–2026); Co-Mentorship Chair (2025–2026); graduate mentor (2024–2026).

## Additional Experience

**Stanford University, Molecular Imaging Instrumentation Lab** Fall 2023
*PhD Rotation Student / Advisor: Prof. Craig Levin*

- Studied conditional GANs for positron emission tomography (PET) image correction.

**Lumafield** May–Aug 2022
*Hardware Research & Development Engineering Co-op*

- Built a multi-detector CT prototype with iterative calibration and OpenCV fusion, doubling capture speed.
- Separately optimized the sharpness–scan-time trade-off, enabling 3× faster scans without quality loss; deployed improvements to customer machines.

**Inertia Product Development** Jan–Apr 2022
*Product Development Intern, Mechatronics*

- Built a customer-facing LiDAR point-cloud visualizer in Python/PyQt with a ROS backend.

**Canadensys** May–Aug 2021
*Aerospace Engineering Intern*

- Developed STM32 firmware and PID control for a high-precision BLDC motor; diagnosed unstable Hall-effect sensor feedback and implemented a software correction that salvaged the PCB design.

**exactEarth (Spire)** Jan–Apr; Sep–Dec 2020
*Software Engineering Co-op*

- Developed autonomous operations software and supported commissioning for EV10 (launched Sept 2020).

## Technical Skills

**Methods:** Transformers, vision-language models, 3D point-cloud processing, multimodal learning.
**Programming & ML:** Python, PyTorch/Lightning, JAX/Flax, C/C++, ROS.
**Training infrastructure:** Distributed training (DDP/FSDP), HPC clusters, Weights & Biases, TensorBoard.

## Selected Coursework

**Vision & ML:** Diffusion & Large Vision Models; 3D/4D Foundation Models; 3D Perception & Reconstruction; Deep Learning for Computer Vision; Virtual Reality.
**Robotics:** Autonomous Mobile Robots; Embedded Systems; Sensors; Automatic Control Systems.